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Impulsivity and aggression: A meta-analysis using the UPPS model of impulsivity☆

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ABSTRACT

Trait impulsivity has long been proposed to play a role in aggression, but the results across studies have been mixed. One possible explanation for the mixed results is that impulsivity is a multifaceted construct and some, but not all, facets are related to aggression. The goal of the current meta-analysis was to determine the relation between the different facets of impulsivity (i.e., negative urgency, positive urgency, lack of premeditation, lack of perseverance, and sensation seeking) and aggression. The results from 93 papers with 105 unique samples (N = 36, 215) showed significant and small-to-medium correlations between each facet of impulsivity and aggression across several different forms of aggression, with more impulsivity being associated with more aggression. Moreover, negative urgency (r = 0.24, 95% [0.18, 0.29]), positive urgency (r = 0.34, 95% [0.19, 0.44]), and lack of premeditation (r = 0.23, 95% [0.20, 0.26]) had significantly stronger associations with aggression than the other scales (rs < 0.18). Two-stage meta-analytic structural equation modeling showed that these effects were not due to overlap among facets of impulsivity. These results help advance the field of aggression research by clarifying the role of impulsivity and may be of interest to researchers and practitioners in several disciplines.

1. Introduction

Aggression and violence are prevalent and costly public health problems that have negative consequences on victims (e.g., injury or death) and perpetrators (e.g., incarceration). Decades of research in anthropology (e.g., Fry, 1998), neuroscience (Gollan, Lee, & Coccaro, 2005; Rosell & Siever, 2015; Verona & Bresin, 2015), psychiatry (Coccaro, 2012; Vitiello & Stoff, 1997), and psychology (Anderson & Bushman, 2002; Berkowitz, 1989; Dodge, 1991; Finkel & Hall, 2018; Wilkowski & Robinson, 2010), among other disciplines has provided some insights into the etiology of aggression; however, more work is needed to identify individual differences that exacerbate or attenuate the likelihood of responding aggressively in order to develop effective prevention and intervention strategies to reduce the public health burden of aggression. Trait impulsivity has long been proposed to play a role in aggression, and there is empirical support for this prediction (e.g., Miller, Zeichner, & Wilson, 2012; Weiss, Connolly, Gratz, & Tull, 2017). Because impulsivity is a multifaceted construct, what remains to be elucidated is whether some, but not all, facets of impulsivity are related to aggression and whether aggression is more strongly related to some aspects of impulsivity than others. The goal of this paper was to conduct a meta-analysis of the relation between impulsivity and aggression using a unifying multifaceted model of trait impulsivity (i.e., the UPPS model; Cyders et al., 2007; Whiteside & Lynam, 2001) to determine the association between unique aspects of impulsivity and aggression.

1.1. A unifying structure for impulsivity

Research on trait impulsivity has a long history in psychology and has been a focus of personality, clinical, and developmental psychologist (Barratt, 1993; Eysenck & Eysenck, 1985; Rothbart, 2007; Whiteside & Lynam, 2001). Over the decades of research, several scales have been created to assess impulsivity (Dickman, 1990; Eysenck, Pearson, Easting, & Allsopp, 1985; Patton, Stanford, & Barratt, 1995; Zuckerman, 1994). These scales, which came from different theoretical backgrounds and areas in psychology, had some conceptual and empirical overlap, but also areas of divergence, which led some to believe that impulsivity was not a unitary construct. In order to clarify the structure of trait impulsivity, Whiteside and Lynam (2001) conducted a factor analysis across several commonly used impulsivity scales. Their results showed a four-factor structure: (negative) urgency, lack of premeditation, lack of perseverance, and sensation seeking. A fifth factor, positive urgency, was later added by Cyders et al. (2007; Cyders &

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Smith, 2008).

Negative urgency refers to an individual's tendency to engage in rash action when feeling negative emotions, whereas positive urgency refers to an individual's tendency to engage in impulsive behavior when feeling positive emotions. Both urgency traits are related to neuroticism in the Big 5 model of personality (Cyders & Smith, 2008). Lack of premeditation refers to an individual's propensity to act without planning. Lack of perseverance indicates an individual's tendency to leave tasks unfinished or to give up. These two factors are related to conscientiousness in the Big 5 (Cyders & Smith, 2008). The final factor, sensation seeking, indicates an individual's engagement in thrill-seeking behaviors and is related to extraversion (Cyders & Smith, 2008).

These studies by Whiteside and Lynam (2001) and Cyders et al. (2007), among others, are important in many ways. First, they provided a unifying framework to organize the multiple facets of impulsivity, which can be used to make predictions about how different facets of impulsivity might be related to different behaviors. For example, a meta-analysis found that negative urgency is more strongly related to nonsuicidal self-injury than lack of perseverance (Hamza, Willoughby, & Heffer, 2015). Similar differential correlations have been found for alcohol and marijuana use (Coskunpinar, Dir, & Cyders, 2013; VanderVeen, Hershberger, & Cyders, 2016). Another advantage of the UPPS model, particularly for meta-analysis is that because it was designed from other impulsivity measures, it is possible to place studies that did not use the UPPS scales into the factor structure. This is particularly advantageous for meta-analysts who wish to include studies prior to 2001.

1.2. Definition of aggression

Throughout time researchers have used several definitions of aggression (e.g., Buss, 1961; Dollard, Doob, Miller, Mowrer, & Sears, 1939; Parrott & Giancola, 2007; Zillmann, 1979). This paper uses the definition put forth by Parrott & Giancola, 2007 among others (e.g., Anderson & Bushman, 2002; Baron & Richardson, 1994) that aggression is a behavior direct toward another with the goal of causing physical or psychological harm that the target wants to avoid. Aggression then is distinct from anger, which is an emotion often associated with aggression, and hostility, which are thoughts associated with aggression (Parrott & Giancola, 2007). Similar to impulsivity, aggression is a heterogeneous construct and theorist has suggested several ways to divide aggression into more homogeneous categories (e.g., direct versus indirect; active versus passive; Gollan et al., 2005; Lesch & Merschdorf, 2000; Parrott & Giancola, 2007).

Based on the available studies, I focused on two dimensions: form and function. The form of aggression refers to the method that is used to deliver harm to another (Anderson & Bushman, 2002; Verona, Sadeh, Case, Reed II, & Bhattacharjee, 2008). Common methods include physical (e.g., punching), verbal (e.g., insulting), relational (spreading rumors), and sexual (insisting that a partner engage in sexual acts; Buss & Perry, 1992; Crick, Ostrov, & Werner, 2006; Straus, Hamby, Boney-McCoy, & Sugarman, 1996; Verona et al., 2008). Different forms of aggression tend to have medium-to-large size correlations with each other (e.g., Evans et al. 2018; Verona et al., 2008), suggesting that people who engage in one form of aggression are likely to engage in multiple forms of aggression. Some scales have unique subscales for each form of aggression (Buss & Perry, 1992; Raine et al., 2006; Straus et al., 1996), whereas other scales consider multiple forms together to create a general aggression score (e.g., Coccaro et al., 1997).

The functions, or purposes, of aggression, are usually divided into proactive (sometimes premeditated) and reactive (also known as impulsive or hostile; Crick & Dodge, 1996; Raine et al., 2006). Proactive aggression is where the physical or psychological pain inflicted is a means to an end. That is, aggression (e.g., hitting someone) is used to obtain some other outcome (e.g., money). Reactive aggression is aggression, where harming the target is the ultimate goal, and this is in response to some form of real or perceived provocation. Some have called into question the utility of the reactive/proactive distinction given that acts of aggression fit neatly into one category (Barratt & Slaughter, 1998; Bushman & Anderson, 2001). In line with this, meta-analytic research has shown a large correlation between reactive and proactive aggression (Polman, de Castro, Koops, van Boxtel, & Merk, 2007). Moreover, many aggression measures do not make the distinction between reactive and proactive aggression. Despite this overlap, there is evidence of unique associations when adjusting for overlap between reactive and proactive aggression (e.g., Miller et al., 2012). Furthermore, there is theoretical overlap between aspects of impulsivity and reactive and proactive aggression; therefore, it seemed useful to examine in this meta-analysis.

1.3. Impulsivity and theories of aggression

Across theories of aggression, there are at least two common aspects, reactivity, and failure to think ahead that are proposed to play a critical role in aggressive behavior(Anderson & Bushman, 2002; Berkowitz, 1989; Verona & Bresin, 2015; Wilkowski & Robinson, 2010). For instance, I^3, which is a meta-theory of aggression, suggests that aggression is a function of three processes or forces: instigation, impellance, and inhibition (Finkel & Hall, 2018). Instigators are situational factors that create the possibility of an aggressive response (e.g., being cut-off in traffic). Impellors are dispositional (or situational) factors that increase the urge to respond aggressively (i.e., reactivity) to the instigator (e.g., generally being an aggressive driver). Inhibitors are factors that help the individual withhold the aggressive response (i.e., inhibit a prepotent response) in the face of urges (e.g., the presence of a police officer) and disinhibitors are factors that increase the likelihood of an aggressive response (e.g., alcohol intoxication). This framework may help make predictions for which aspects of impulsivity might be related to aggression.

Negative urgency (and to a lesser extent positive urgency) is a trait that could increase the risk of aggression in two ways. First, urgency may function as an impellor and increase reactivity to aggression-instigating situations. That is, for the same aggression-instigating event (e.g., being cut off in traffic) individuals high in negative urgency, may have a stronger anger reaction than individuals low in negative urgency. This fits with the idea that negative urgency is closely related to neuroticism (Cyders & Smith, 2008). Second, urgency may also be a disinhibitor, making it easier for aggressive responses to occur when provoked. By definition, urgency is acting rash in the face of affect. This could be easily translated into acting aggressively in the face of anger. Along these lines, studies have found significant positive associations between negative and positive urgency and aggression (Derefinko, DeWall, Metze, Walsh, & Lynam, 2011; Leone, Crane, Parrott, & Eckhardt, 2016; Miller et al., 2012).

Lack of premeditation could increase aggression primarily as a disinhibitor. On the one hand, people who have a difficult time planning for the future will likely have difficulty thinking about the future consequences of their aggressive actions (e.g., causing problems in a relationship). This may allow aggressive behavior to occur in the presence of an instigator. On the other hand, people who are high in premeditation may be better able to inhibit aggression when the urge is high because they can think about the long-term repercussions of behaving aggressively. Understanding these long-term consequences may help these individuals quell their urges to aggress. Consistent with this, research has found positive correlations between lack of premeditation and aggression (Derefinko et al., 2011; Leone et al., 2016).

The final two aspects of impulsivity, sensation seeking and lack of perseverance, are presumed to have weak positive associations with aggression. Due to their quest for novelty, individuals high in sensation seeking may be more likely to be in aggression-instigating situations. For instance, someone high (versus low) in sensation seeking might be more likely to provoke someone to “get a rise” out of them, which may result in an aggressive interaction. Lack of perseverance may be related to aggression to the extent that inhibiting an aggressive urge requires persistence; however, lack of perseverance, at least as measures by the UPPS scales, is focused more on task persistence than emotion regulation, as evidenced by its association with conscientiousness (Cyders & Smith, 2008). In line with this, research using the UPPS has found weak and inconsistent correlations between sensation seeking and lack of perseverance and aggression (Derefinko et al., 2011; Latzman & Vaidya, 2013; Leone et al., 2016).

1.4. Moderators

It is possible that different aspects of impulsivity might have stronger (or weaker) relations with different forms and/or functions of aggression. Individual studies have generally found similar correlations between facets of impulsivity and aggression across different forms of aggression (e.g., Carlson, Pritchard, & Dominelli, 2013; Weiss et al., 2017). Thus, there is little a priori reason to predict different relations among facets of impulsivity and different forms of aggression. Still, an empirical test would be helpful for future theory development. Prior studies have shown similar sized zero-order correlations between different facets of impulsivity and reactive and proactive aggression (Hecht & Latzman, 2015; Miller et al., 2012). When considering the unique variance (i.e., adjusting for overlap between functions), different results have appeared for the distinct facets of impulsivity. Specifically, negative urgency appears to have a significant positive relation to the unique variance of reactive aggression (Hecht & Latzman, 2015; Miller et al., 2012). This is consistent with the intuition that people who tend to engage in impulsive behavior when upset may be more prone to engage in aggression that is reactive. There is also some evidence that lack of premeditation is positively related to the unique variance of proactive aggression (Hecht & Latzman, 2015).

1.5. Current study

The goal of the current meta-analysis was to determine the relation between the different facets of impulsivity, as defined by the UPPS, and aggression. This was done using two-stage meta-analytic structural equation modeling, which allowed for the examination of zero-order relations and relations adjusting for the other facets of impulsivity.[1] Based on prior research and theory (Miller et al., 2012; Weiss et al., 2017; Cyders & Smith, 2008), it was predicted that negative urgency, positive urgency, and lack of premeditation would have significant and small-to-medium zero-order and partial correlations with aggression. No strong predictions were made for sensation seeking or lack of perseverance as results have been fairly inconsistent in prior studies.

In addition to the primary goal, I also conducted moderation analyses to see whether the correlations were similar across different forms and functions of aggression and across different samples. It was assumed that the correlation between different facets of impulsivity and aggression would be similar across different forms of aggression (e.g., physical, verbal). For function, it was predicted that negative and positive urgency would be related to the unique variance of reactive aggression. In addition to form and function, I explored whether sample demographic characteristics (e.g., sample age, sample gender, sample type) were related to the correlations between impulsivity and aggression in the Supplemental Materials.

[1] Caution should be used when interpreting partial correlations, particularly when the nomological network for the unique variance has not been identified. (e.g., Lynam, Hoyle, & Newman, 2006; Sleep, Lynam, Hyatt, & Miller, 2017). Still, partial correlations can be useful as robustness tests to clarify relations among variables (Verona & Miller, 2014).

2. Method

2.1. Literature search

Search terms for impulsivity were derived from prior meta-analyses using the UPPS model (Coskunpinar et al., 2013; Kale, Stautz, & Cooper, 2018). Specifically, I searched the combination of each scale (see Supplemental Table 1) that has been proposed to map onto UPPS with the word aggression. The search yielded 627 unique papers. Fig. 1 shows the number of papers involved in each stage of determining eligibility. During the search, each paper's method section was reviewed to determine whether there was a measure of impulsivity that mapped into the UPPS and a measure of aggression/violence, which yielded 161 eligible papers. Studies were excluded if they were not in English, were a review paper or meta-analyses, did not have an impulsivity scale that fit into the UPPS, or was an experimental study (unless correlations were reported prior to the manipulation). Studies that used suicidality as the aggressive outcome were also excluded as this meta-analysis was focused on aggression directed toward others, not the self.

Seventy-nine studies had enough data to calculate the correlation between one measure of impulsivity and aggression. Corresponding authors of the 77 papers that did not report the relevant correlations were contacted and 14 provided the necessary correlations, leading to a final of 93 papers with 105 unique samples (some papers had more than one sample; Vigil-Colet, Morales-Vives, & Tous, 2008) and a total samples size of 36, 215.

2.2. Study coding

Several study characteristics and sample characteristics were coded. In addition to sample size, the gender, racial/ethnic background, and mean age of the sample were recorded. The country where the data were collected was coded as was the type of sample (undergraduates, community, clinical, forensic, other). Of primary interest were the correlations among the impulsivity scales and between the impulsivity scales and measures of aggression. See Tables 1 and 2 for summary and study level characteristics. Aggression scales were categorized as general (i.e., combing 2 or more forms of aggression; number of effects = k = 103), physical (k = 38), verbal (k = 30), sexual (k = 3), relational (k = 7), or other (k = 6). Self-reported aggression was the most common method (k = 165) followed by informant (k = 11), other (e.g., interview; k = 8), records (k = 2), and behavior (k = 1).

2.3. Data analytic plan

The primary analyses used two-stage meta-analytic structural equation modeling (Cheung, 2014; Cheung & Chan, 2005). In the first stage, a pooled correlation matrix was estimated using maximum likelihood estimation. This provided the meta-analyzed zero-order correlations between the UPPS scales (Negative Urgency, Sensation Seeking, Lack of Premeditation, Lack of Perseverance) and aggression (i.e., general, physical, verbal). A subset of scales for impulsivity and aggression were used due to model fit issues when positive urgency and other forms of aggression, which were only available for a subset of studies, were included. Because there were missing values in the correlation matrices across studies (i.e., not all studies had all measures), a random effects model was used (Cheung, 2014). In the second stage, the pooled matrix was used to fit the structural equation model. The model created a latent aggression variable indicated by general, physical, and verbal aggression. This latent variable was predicted by the four UPPS scales, which were allowed to covary. This allowed for the test of the association between indices of impulsivity and aggression adjusting for overlap with other measures of impulsivity. The variances for the predictors were set to 1 all other variances were estimated. Model fit was evaluated based on the standard structural equation modeling fit

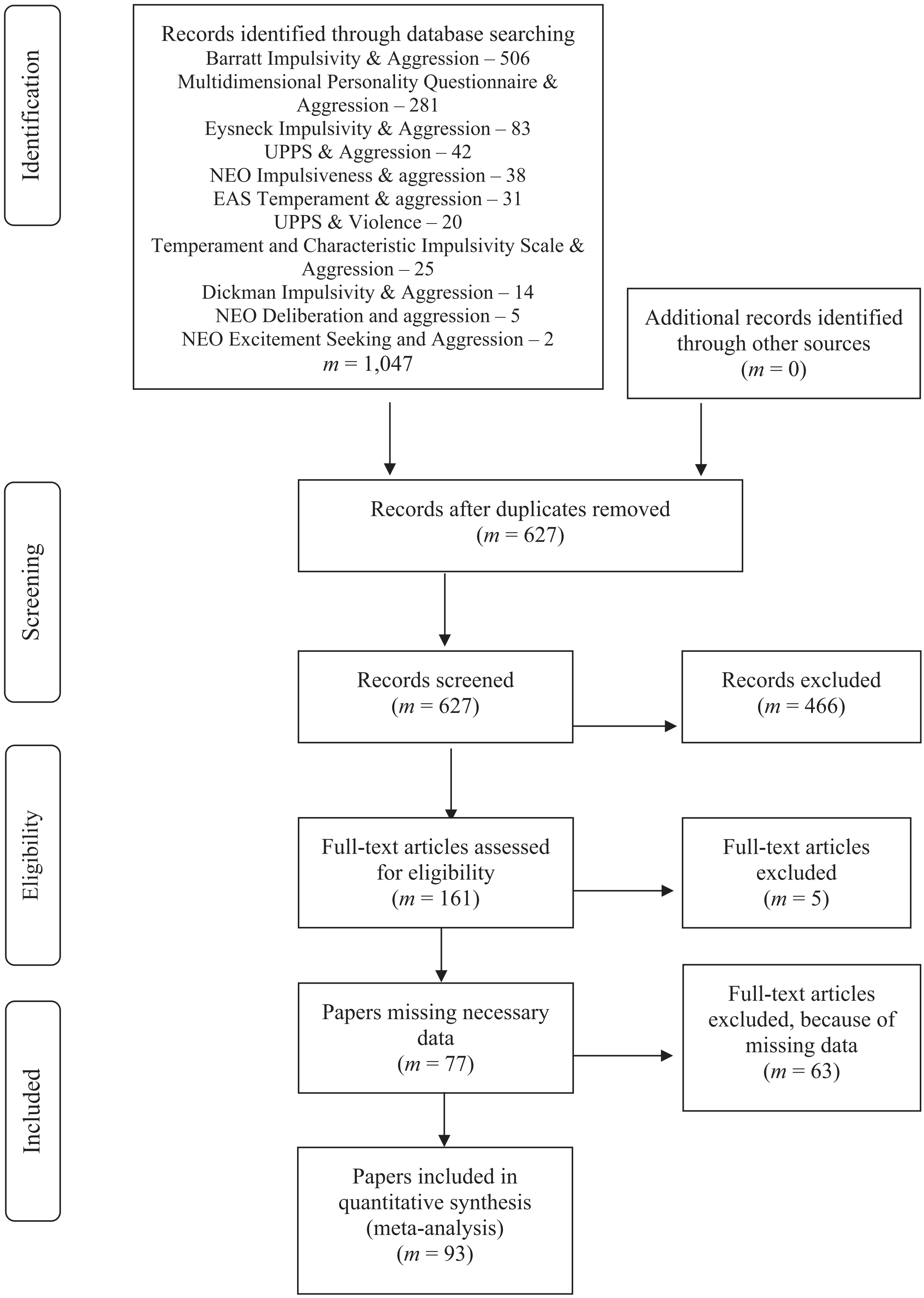

Fig. 1. Flow Diagram of Literature Search and Screening Process (m = number of papers).

Table 1
Study characteristics in aggregate.

	M	SD	Min	Max
N (*m* = 105)	344.90	319.81	33	2295
% Women (*m* = 100)	41.88	28.35	0	100
Race/ethnicity				
% White (*m* = 39)	63.61	19.02	27.5	100
% Black/African American (*m* = 33)	20.75	17.71	2.3	63.3
% Latinx/Hispanic (*m* = 22)	10.28	6.79	1	25
% Asian/Asian American (*m* = 23)	10.18	10.18	0.4	41
% Indigenous (*m* = 4)	4.3	3.68	0.08	9
% More than one Race (*m* = 2)	11.5	4.95	8	15
% Other (*m* = 13)	5.36	4.74	1	16
Age (*m* = 98)	27.46	10.64	11.1	55.6
Year (*m* = 105)	2011	5.45	1994	2019

Country	*m*	%
North America	62	58.49
United States	57	53.77
Canada	4	3.77
Europe	39	36.79
Belgium	1	0.94
Denmark	2	1.89
France	2	1.89
Germany	4	3.77
Greece	1	0.94
Italy	12	11.32
Netherlands	9	8.49
Portugal	1	0.94
Slovakia	1	0.94
Spain	5	4.71
United Kingdom	1	0.94
Asia	3	2.83
China	2	1.89
Korea	1	0.94
Africa/Middle East	2	1.89
Israel	1	0.94
Ghana	1	0.94
Sample Type		
Undergraduates	36	33.96
Community	41	38.67
Clinical	17	16.03
Incarcerated	5	4.72
Other	8	7.54

Note. *m* = number of studies; Indigenous = Native American, First Nation, and/or Pacific Islanders.

indices (e.g., CFI, TLI, and RMSEA). The analyses were conducted with the metaSEM package in R (Cheung, 2015).

In addition to the main analyses, there were two sets of auxiliary analyses. Because so few studies included positive urgency ($m = 14$) and the stage 1 meta-analytic structural equation modes failed to converge, I examined the correlation between positive urgency and aggression using multivariate meta-analysis with robust variance estimation and the small sample adjustment (Tipton, 2015). This method was chosen because it adjusts for an unknown dependency when multiple effects (k) come from one study (m; Tipton, 2015). These models were fit with the *robumeta* package in R (Tanner-Smith, Tipton, & Polanin, 2016). A similar strategy was used to examine the correlation between impulsivity and reactive and proactive aggression, which were reported in six studies. These analyses were run on z-transformed correlations, but correlations were transformed to their original units for the text. For proactive and reactive aggression, I also calculated correlations between impulsivity and one function of aggression (e.g., reactive) with the other function (e.g., proactive) partialed out using the meta-analyzed correlation between functions of aggression. Additional analyses presented in the Supplemental Materials show examination of whether the correlation between facets of impulsivity and aggression varied as a function of sample characteristics (i.e., sample age, sample gender composition, sample type) and the effect sizes for each different measure of impulsivity.

3. Results

3.1. Study characteristics

Table 1 displays the aggregate characteristics of the studies included in the meta-analysis and Table 2 displays this information by study. The average sample size across the samples was 345, which according to G*Power (Faul, Erdfelder, Lang, & Buchner, 2007) would give the power of 0.45 to detect small correlations as defined by Cohen ($r = 0.10$; 1988) and 0.99 power for medium and large correlations ($r > 0.30$). The samples were fairly young and contained more men than women. The studies that reported race and ethnicity showed a breakdown similar to the United States census numbers (United States Census Bureau, 2017); however, only 37.5% of studies reported race/ethnicity. Over half of the studies were conducted in the United States, and about a third were conducted in Europe. Approximately a third of the samples were undergraduates, a third were community members, and a third were clinical (e.g., inpatient), incarcerated, or other (e.g., veterans, refugees).

3.2. Primary analyses

3.2.1. Meta-analytic structural equation modeling stage 1

Table 3 displays the pooled correlation matrix. As predicted, the results showed that all but one of the correlations between impulsivity and aggression are significantly different from zero, with the one exception being lack of perseverance and verbal aggression. All correlations indicated that more impulsivity was related to more aggression. The effects were small-to-medium in size for negative urgency and lack of premeditation ($r > 0.23$) and small for sensation seeking and lack of perseverance ($r > 0.05$). Moreover, the effects were significantly larger for negative urgency and lack of premeditation than sensation seeking and lack of perseverance, as indicated by non-overlapping confidence intervals. The effect sizes were similar across general, physical, and verbal aggression.

3.2.2. Meta-analytic structural equation modeling stage 2

The model fit indices suggested a good fit to the data, CFI = 0.995, TLI = 0.998, RMSA = 0.005, 95% CI [0.000, 0.009], χ^2 (8) = 14.57, $p = .068$. As shown in Fig. 2, the three indicators of aggression loaded strongly on the latent aggression factor (> 0.60). The path coefficients were significant for three of the four indices of impulsivity: negative urgency, sensation seeking, and lack of premeditation. As with the zero-order correlations, the associations for negative urgency ($\beta = 0.27$ [0.21, 0.34]) and lack of premeditation ($\beta = 0.23$ [0.18, 0.31]) were significantly larger than for sensation seeking ($\beta = 0.09$ [0.03, 0.15]), with small-to-medium effects for negative urgency and lack of premeditation a small effect for sensation seeking. Although caution should be taken in interpreting these relations because the nomological network for the partial facets of impulsivity are poorly characterized, these results suggest that the significant zero-order correlations are not due purely to overlapping variance across facets of impulsivity.

3.3. Additional analyses

3.3.1. Positive urgency

As with negative urgency, there was a significant positive correlation between positive urgency and general aggression, $r = 0.34$, 95% [0.19, 0.44], $k = 11$, $m = 8$. There was also a significant correlation for the three physical aggression studies, $r = 0.26$, 95% [0.14, 0.38], $k = 4$, $m = 4$. The one study that reported the correlation between positive urgency and verbal aggression found a non-significant correlation similar in size to those for general and physical aggression, $r = 0.25$, 95% [−0.20, 0.69], $k = 1$, $m = 1$.

Table 2
Study characteristics by study.

Authors	Year	*N*	%Women	Age	Country	Sample Type	Impulsivity Measure(s)	UPPS facets	Aggression measure(s)
Adjorlolo et al.	2017	363	59.00	17	Ghana	Community	Barratt Impulsivity Scale	NU PRE	Violent Delinquency, Buss-Perry Aggression Physical Aggression, Buss-Perry Aggression Verbal Aggression
Ammerman et al.	2015	2295	61.00	20.94	United States	Undergraduates	UPPS	NU SS PRE PER	Lifetime History of Aggression
Anguiano-Carrasco & Vigil-Colet	2011	935	53.00	23.47	Spain	Undergraduates	Dickman Impulsivity Scale	SS	Indirect Aggression Scales - Aggressor
Augsburger et al.	2017	94	15.00	30.1	Germany	Other: Refugees	UPPS	NU SS PRE PER	Appetitive and Facilitative Aggression Scale Appetitive Scale, Appetitive and Facilitative Aggression Scale Reactive Scale
Bailey & Ostrov	2008	165	50.00	19.05	United States	Undergraduates	Barratt Impulsivity Scale	PRE	Self-report of Aggression and Social Behavior Measure - proactive relational aggression Self Report of Aggression and Social Behavior Measure -reactive relational aggression Self Report of Aggression and Social Behavior Measure -proactive physical aggression Self Report of Aggression and Social Behavior Measure -reactive physical aggression
Barratt et al.	1999	216	80.00	27.38	United States	Undergraduates	Barratt Impulsivity Scale	NU PRE	Aggressive Acts Questionnaire - Impulsive, Aggression Aggressive Acts Questionnaire - Premeditated Aggression
Berdoulat et al.	2013	455	30.00	32.13	French	Community	UPPS	NU SS PRE PER	Driving Behavior Questionnaire Transgression Subscale - Aggressive Transgressions
Bousardt et al.	2015	74	22.00	35	Netherlands	Clinical	UPPS	NU PU SS PRE PER	Social Dysfunction and Aggression Scale
Bresin & Verona	2016	114	54.00	19.65	United States	Undergraduates	UPPS	NU	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
Bresin & Verona	2016	142	51.00	19.28	United States	Undergraduates	UPPS	NU	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
Carlson et al.	2013	282	50.00	20.78	Canada	Undergraduates	UPPS	NU PU SS PRE PER	Buss-Perry Aggression Questionnaire - Physical and Verbal
Carrasco et al.	2006	868	0.00		Canada	Community	Eysenck I-7 Scale	SS PRE	Self-reported Antisocial behaviors Questionnaire - Aggressive subscale
Carver et al.	2014	368	65.22	18.71	United States	Undergraduates	UPPS	NU PU PER	Buss-Perry Aggression Questionnaire - Physical Aggression, Buss-Perry Aggression Questionnaire - Verbal Aggression
Chen et al.	2013	651	59.00	26.13	China	Undergraduates	Barratt Impulsivity Scale	NU PRE	Impulsive/Premeditated Aggression Scale - Impulsive Aggression, Impulsive/Premeditated Aggression Scale - Premeditated Aggression
Cima et al.	2013	214			Netherlands	Other: Incarcerated and Community	Barratt Impulsivity Scale	NU PRE	The Reactive Proactive Questionnaire - Reactive Scale, The Reactive Proactive Questionnaire - Proactive Scale
Claes et al.	2009	399	45.10	31.37	Belgium	Clinical	Dickman Impulsivity Scale	SS PRE	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression

(*continued on next page*)

Table 2 (*continued*)

Authors	Year	*N*	%Women	Age	Country	Sample Type	Impulsivity Measure(s)	UPPS facets	Aggression measure(s)
Collins et al.	2012	225	42.00	26.55	United Kindgom	Community	Brief Sensation Seeking Scale	SS	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
Collison et al.	2018	280	65.00		United States	Community	UPPS	NU PU SS PRE PER	Reactive and proactive aggression questionnaire - Reactive, Reactive and proactive aggression questionnaire - Proactive, Revised Self-report of Aggression and Social Behavior - Reactive Peer Aggression, Revised Self-Report of Aggression and Social Behavior - Proactive Peer Aggression, Revised Self-report of Aggression and Social Behavior - Relationship Aggression
Comai et al.	2016	361	0.00	44.36	Canada	Incarcerated	Barratt Impulsivity Scale	NU PRE	MacArthur Community Violence Instrument
Constantinou et al.	2011	352	31.00	20.29	Greece	Other: Undergraduates and National Guard Sevice Members	Barratt Impulsivity Scale, Sensation Seeking Scale	NU SS PRE PER	Driving Behavior Questionnaire - Aggression Scale
Cosi et al.	2011	563	54.00	11.24	Spain	Community	Barratt Impulsivity Scale	NU PRE	Proactive/Reactive Aggression Questionnaire for Teachers - Reactive, Proactive/Reactive Aggression Questionnaire for Teachers - Proactive
Critchfield et al.	2004	92	92.00	30.75	United States	Clinical	MPQ Control Scale	PRE	Anger, Irritability, Assault Questionnaire - Direct Assault Anger, Irritability, Assault Questionnaire - Verbal Assault Anger, Irritability, Assault Questionnaire - Indirect Assault Anger, Irritability, Assault Questionnaire - Irritability Overt Aggression Scale, Modified for Outpatient - Aggression Overt Aggression Scale
Crossman	1995	480	0.00	19.2	United States	Undergraduates	Barratt Impulsivity Scale	NU PRE	Sexual Experiences Survey
Cunha-Bang et al.	2013	94	36.00	47	Denmark	Community	NEO Impulsiveness Scale, Barratt Impulsivity Scale	NU PRE	Buss-Perry Aggression Questionnaire - Total
Cunha-Bang et al.	2016	61	22.95	33.8	Denmark	Community	Barratt Impulsivity Scale	NU PRE	Buss-Perry Aggression Questionnaire - Physical Aggression, Buss-Perry Aggression Questionnaire - Verbal Aggression
Denny & Siemer	2012	112	52.00	19.4	United States	Undergraduates	UPPS	NU PU	Buss-Perry Aggression Questionnaire - Total
Derefinko et al.	2011	131		19.7	United States	Undergraduates	UPPS	NU SS PRE PER	Crime and Analogous Behavior Scale - General, Violent Behavior Crime and Analogous Behavior Scale - Intimate Partner Violence
de Schutter et al.	2016	84	50.00	37.5	Netherlands	Other: Emergency Departments and Protected Housing	Dickman Impulsivity Scale	PRE	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
Dogan et al.	2010	553	52.00	15	United States	Community	NEO Impulsiveness Scale, NEO-PI-R Excitement Seeking Scale	NU SS	Hostility Inventory
Burt & Donnellan	2008	292	51.00	19	United States	Undergraduates	MPQ Control Scale	PRE	Adult Self-Report
Burt & Donnellan	2008	137	0.00	19	United States	Undergraduates	MPQ Control Scale	PRE	Adult Self-Report- Aggressive Antisocial Behavior, Buss-Perry Aggression Questionnaire - Physical and Verbal
Duran-Bonavila et al.	2017	532	53.00	14.75	Spain	Community	Barratt Impulsivity Scale	NU PRE	The Indirect-Direct Aggression Questionnaire - Physical Aggression, The Indirect-Direct Aggression Questionnaire- Verbal Aggression, The Indirect-Direct Aggression Questionnaire - Indirect Aggression
Eigenhuis et al.	2015	1055	52.00	45.83	Netherlands	Community	MPQ Control Scale	PRE	MPQ - Aggression
Eigenhuis et al.	2015	1055	52.00	45.6	United States	Community	MPQ Control Scale	PRE	MPQ - Aggression
Eigenhuis et al.	2015	410	69.00	19.69	Netherlands	Undergraduates	MPQ Control Scale	PRE	MPQ - Aggression
Eigenhuis et al.	2015	410	69.00	18.78	United States	Undergraduates	MPQ Control Scale	PRE	MPQ - Aggression

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Table 2 (*continued*)

Authors	Year	*N*	%Women	Age	Country	Sample Type	Impulsivity Measure(s)	UPPS facets	Aggression measure(s)
Flory et al.	2006	351	56.00	42	United States	Community	Sensation Seeking Scale, Barratt Impulsivity Scale	SS PRE PER	Buss-Perry Aggression Questionnaire - Total
Flory et al.	2006	70	45.00	35	United States	Clinical	Sensation Seeking Scale, Barratt Impulsivity Scale	SS PRE PER	Buss-Durkee Hostility Inventory - Total Score
Fossati et al.	2007	461	63.30	33.9	Italy	Clinical	Dickman Impulsivity Scale, BP - Perseverance Scale	PRE PER	Buss-Perry Aggression Questionnaire
Fossati et al.	2004	747	64.50	22.9	Italy	Undergraduates	Barratt Impulsivity Scale	NU PRE	Buss-Durkee Hostility Inventory -Physical Aggression, Buss-Durkee Hostility Inventory - Indirect Aggression, Buss-Durkee Hostility Inventory - Verbal Aggression
Gagnon & Rochat	2017	177	45.00	25.21	Canada	Undergraduates	UPPS	NU PU SS PRE PER	Reactive-Proactive Aggression Questionnaire - Reactive Scale
Gansler et al.	2009	41	13.00	40.12	United States	Clinical	Barratt Impulsivity Scale	PRE	History of Aggression Scale-Revised -Aggression
Gansler et al.	2011	36	0.00	39.47	United States	Clinical	Barratt Impulsivity Scale	NU PRE	Lifetime History of Aggression
Garofalo et al.	2018	221	0.00	40.9	Italy	Incarcerated	Barratt Impulsivity Scale	NU PRE	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
Garofalo et al.	2018	245	0.00	38.9	Italy	Community	Barratt Impulsivity Scale	NU PRE	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
Gauthier et al.	2009	56	39.00	14.37	United States	Clinical	NEO Impulsiveness Scale, NEO-PI-R Excitement Seeking Scale, NEO Deliberation Scale, NEO Self-Discipline Scale	NU SS PRE PER	Impulsive/Premeditated Aggression Scale - Impulsive, Impulsive/Premeditated Aggression Scale - Premeditated
Grimaldi et al.	2014	245	65.70	18.9	United States	Undergraduates	UPPS	NU PU	Indirect Aggression Scale - Aggressor
Harkness et al.	1995	232			United States	Undergraduates	MPQ Control Scale	PRE	MPQ - Aggression
Hecht & Latzman	2015	384	56.80	20.92	United States	Undergraduates	UPPS	NU PU SS PRE PER	Reactive Proactive Aggression Questionnaire- Reactive, Proactive Aggression Questionnaire- Proactive
Heinz et al.	2015	133	0.00	52.52	United States	Other: Veterans	MPQ Control Scale	PRE	The Revised Conflict Tactics Scale - Physical Aggression
Hicks et al.	2008	1329		17.5	United States	Community	MPQ Control Scale	PRE	MPQ - Aggression
Jones	2017	277	62.34	21	United States	Undergraduates	UPPS	NU SS PRE PER	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
Kahler et al.	2010	296	67.90	38.8	United States	Community	MPQ Control Scale	PRE	MPQ - Aggression
Kennealy et al.	2007	226	100.00	31.9	United States	Incarcerated	MPQ Control Scale	PRE	MPQ - Aggression
Kovácsová et al.	2016	578	37.20	32.8	Slovakia	Community	Dickman's Impulsivity Scale	PRE	Driving Aggressive Warnings Driving Hostile Aggression and Revenge
Krakowski & Czobor	2018	74	21.00	38.5	United States	Clinical	Barratt Impulsivity Scale	NU PRE	Lifetime History of Aggression

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Table 2 (*continued*)

Authors	Year	*N*	%Women	Age	Country	Sample Type	Impulsivity Measure(s)	UPPS facets	Aggression measure(s)
Krakowski & Czobor	2018	70	19.00	38	United States	Community	Barratt Impulsivity Scale	NU PRE	Lifetime History of Aggression
Latzman & Vaidya	2013	933	73.50	20.81	United States	Undergraduates	UPPS	PRE PER	Reactive Proactive Aggression Questionnaire - Reactive, Reactive Proactive Aggression Questionnaire - Proactive
Latzman et al.	2011	413	71.10	18.35	United States	Undergraduates	NEO Deliberation Scale, NEO Self-Discipline Scale	PRE PER	Reactive Proactive Aggression Questionnaire - Reactive, Reactive Proactive Aggression Questionnaire - Proactive
Lavner et al.	2017	172	100.00	27.6	United States	Community	Dickman's Impulsivity Scale	PRE	Conflict Tactics Questionnaire - Verbal Subscale
Lavner et al.	2017	172	0.00	27.6	United States	Community	Dickman's Impulsivity Scale	PRE	Conflict Tactics Questionnaire - Verbal Subscale
Lejoyeux et al.	2013	100	58.00	41.9	France	Clinical	Barratt Impulsivity Scale, Sensation Seeking Scale	NU SS PRE PER	Overt Aggression Scale
Leone et al.	2016	578	50.00	32.54	United States	Community	UPPS	NU PU SS PRE PER	Conflict Tactics Questionnaire - Physical Subscale
Lievaart et al.	2016	764	75.00	20.4	Netherlands	Undergraduates	Barratt Impulsivity Scale	NU PRE	STAXI-2 Anger Out
Lievaart et al.	2016	226	65.40	44.45	Netherlands	Clinical	Barratt Impulsivity Scale	NU PRE	STAXI-2 Anger out
Lynam & Miller	2004	211	59.00		United States	Undergraduates	UPPS	NU SS PRE PER	% of Aggressive Responses to Vignettes Aggression Enacted on Vignettes
Maneiro et al.	2017	575	54.00	15.94	Spain	Community	UPPS	NU PU SS PRE PER	Antisocial Behaviour Questionnaire - Aggression
Martinotti et al.	2008	385	57.70	32.59	Italy	Community	Temperament and Character Inventory	PRE	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
McDermott et al.	2008	238	14.00	46.6	United States	Clinical	Barratt Impulsivity Scale	NU PRE	Special Incident Report - Impulsive Aggression, Special Incident Report - Predatory Aggression, Special Incident Report - Psychotic Aggression
McGrath et al.	2012	192	100.00	42.9	United States	Community	Sensation Seeking Scale	SS	Couples Conflicts and Problem-Solving Strategies Questionnaire - Physical, Couples Conflicts and Problem-Solving Strategies Questionnaire - Verbal, Child Rearing Issues: Discipline subsection of the Hetherington Discipline Scale
Miller et al.	2003	481		21	United States	Community	NEO Impulsiveness Scale, NEO-PI-R Excitement Seeking Scale, NEO Deliberation Scale, NEO Self-Discipline Scale	NU SS PRE PER	Conflict Tactics Scale
Miller et al.	2012	116	44.00	19.16	United States	Undergraduates	UPPS	NU PU SS PRE PER	Reactive Proactive Aggression Questionnaire - Reactive, Reactive Proactive Aggression Questionnaire - Proactive, Self-report Measure of Aggression - Relational

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Table 2 (*continued*)

Authors	Year	*N*	%Women	Age	Country	Sample Type	Impulsivity Measure(s)	UPPS facets	Aggression measure(s)
Miller et al.	2013	368	22.00	35.3	United States	Community	MPQ Control Scale	PRE	MPQ - Aggression
Pawliczek et al.	2013	33	0.00	22.27	Germany	Undergraduates	Barratt Impulsivity Scale	NU PRE	Buss-Perry Aggression-Total Score
Pechorro et al.	2016	782	52.25	15.87	Portugal	Community	Barratt Impulsivity Scale	NU PRE	Reactive-Proactive Aggression Questionnaire
Puhalla et al.	2016	191	55.49	28.41	United States	Other: Undergraduates and Outpatient	UPPS	NU PU SS PRE PER	STAXI Anger Expression-out
Roozen et al.	2011	80	27.00	37	Netherlands	Community	Dickman's Impulsivity Scale	SS PRE	Buss-Perry Aggression Questionnaire - Physical Aggression, Buss-Perry Aggression Questionnaire - Verbal Aggression
Sadeh et al.	2009	229	57.60	14.2	United States	Community	MPQ Control Scale	PRE	MPQ - Aggression
Schiffer et al.	2011	51	0.00	36.95	Germany	Other: Community and Clinical	Barratt Impulsivity Scale	NU PRE	People Who had Been Convicted of 3 or more Violent Crimes
Seibert et al.	2010	137	42.33	19.33	United States	Undergraduates	UPPS	NU PU SS PRE PER	Taylor Aggression - Average of Intensity, Duration, and Frequency
Seroczynski et al.	1999	628	0.00	43.97	United States	Community	Barratt Impulsivity Scale	PRE	Buss-Durkee Hostility Inventory - Direct Aggression, Buss-Durkee Hostility Inventory- Verbal Assault, Buss-Durkee Hostility Inventory - Indirect Aggression
Settles et al.	2012	418	75.00	18.2	United States	Undergraduates	UPPS	NU PRE PER	Risky Behavior Scale - Fights Item
Shechory-Bitton & Kamel	2014	120	100.00	16.27	Israel	Community	MPQ Control Scale	PRE	MPQ - Aggression
Shoal et al.	2003	314	0.00	11.37	United States	Community	MPQ Control Scale	PRE	MPQ- Aggression, Child Behavior Checklist - Aggressive Behavior
Sideli et al.	2017	652	0.00		Italy	Community	Sensation Seeking Scale	SS	Zuckerman-Kuhlman Personality Questionnaire Aggression Scale
Soloff et al.	2003	121	53.72	25.5	United States	Community	Barratt Impulsivity Scale	PRE	Lifetime History of Aggression Buss-Durkee Hostility Inventory Total
Stanford et al.	1995	214	72.00	25.09	United States	Undergraduates	Barratt Impulsivity Scale	PRE	Buss-Durkee Hostility Inventory - Indirect Aggression, Buss-Durkee Hostility Inventory - Verbal Aggression, Anger Attack Questionnaire - Impulsive Aggression
Stetler et al.	2014	89	0.00	31.81	United States	Incarcerated	Barratt Impulsivity Scale	NU PRE	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
Surís et al.	2005	474	9.10	55.6	United States	Other: Veterans	Barratt Impulsivity Scale	NU PRE	Overt Aggression Scale, Modified Verbal Overt Aggression Scale, Modified - Object Assault Overt Aggression Scale, Modified - Other Assault, Buss-Perry Aggression Questionnaire Physical Aggression Buss-Perry Aggression Questionnaire Verbal Aggression
Thompson et al.	2006	40	25.00	16.43	United States	Community	Eysenck I-7 Scale	PRE	Peak Aggressive Behavior Rating Scale, Child Behavior Checklist Youth Self-Report
Velotti et al.	2016	617	45.80	36.88	Italy	Community	Barratt Impulsivity Scale	NU PRE	Buss-Perry Aggression Questionnaire - Total
Velotti et al.	2016	257	50.80	43.89	Italy	Clinical	Barratt Impulsivity Scale	NU PRE	Buss-Perry Aggression Questionnaire - Total
Verschuere et al.	2012	57	0.00	16.75	Netherlands	Community	Barratt Impulsivity Scale	NU PRE	Buss-Perry Aggression Questionnaire - Physical Aggression, Buss-Perry Aggression Questionnaire - Verbal Aggression, Reactive-Proactive Aggression Questionnaire - Reactive, Reactive-Proactive Aggression Questionnaire Proactive

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Table 2 (*continued*)

Authors	Year	*N*	%Women	Age	Country	Sample Type	Impulsivity Measure(s)	UPPS facets	Aggression measure(s)
Vigil-Colet et al.	2012	538	60.00	23.65	Spain	Undergraduates	Dickman's Impulsivity Scale	PRE	Indirect Aggression Scale, Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
Vigil-Colet & Raga	2004	84	80.00	23.7	Italy	Undergraduates	Dickman's Impulsivity Scale	SS PRE	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
Vigil-Colet et al.	2008	216	79.00	22.97	Italy	Undergraduates	Dickman's Impulsivity Scale	SS PRE	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
Vigil-Colet et al.	2008	241	55.00	14.2	Italy	Community	Dickman's Impulsivity Scale	SS PRE	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
Vigil-Colet et al.	2008	323	35.00	34.2	Italy	Community	Dickman's Impulsivity Scale	SS PRE	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
Voller et al.	2010	521	0.00	20.24	United States	Undergraduates	NEO Impulsiveness Scale, NEO-PI-R Excitement Seeking Scale, NEO Deliberation Scale, NEO Self-Discipline Scale	NU SS PRE PER	Sexual Experiences Survey
Wacker et al.	2013	201	0.00	23.8	Germany	Undergraduates	MPQ Control Scale	PRE	MPQ - Aggression
Wang & Diamond	1999	385	0.00		United States	Clinical	Barratt Impulsivity Scale	NU PRE	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
Weiss et al.	2017	92	25.00	40.25	United States	Clinical	UPPS	NU SS PRE PER	Buss-Perry Aggression Questionnaire Physical Aggression, Buss-Perry Aggression Questionnaire Verbal Aggression
Yoo et al.	2006	516	51.00	11.1	Korea	Community	Temperament and Character Inventory	PRE	CBCL Aggression
Zhang et al.	2015	184	0.46	14.2	United States	Community	MPQ Control Scale	PRE	MPQ - Aggression, CBCL- Aggression
Zhou et al.	2014	323	0.00	16.6	China	Incarcerated	Barratt Impulsivity Scale	NU PRE	Violent versus Non-violent Offence

Note. NU = Negative Urgency; PU = Positive Urgency; SS = Sensation Seeking; PRE = Lack of Premeditation, PER = Lack of Perseverance.

Table 3
Meta-analyzed correlation matrix for impulsivity and primary aggression outcomes.

	1.	2.	3.	4.	5.	6.	7.
1. Negative urgency	–						
2. Sensation seeking	0.14 [0.10, 0.18]	–					
3. Lack of premeditation	0.38 [0.32, 0.44]	0.16 [0.08, 0.24]	–				
4. Lack of perseverance	0.29 [0.20, 0.38]	−0.00 [−0.08, 0.08]	0.45 [0.36, 0.52]	–			
5. General aggression	0.24 [0.18, 0.29]	0.11 [0.04, 0.18]	0.23 [0.20, 0.26]	0.18 [0.12, 0.24]	–		
6. Physical aggression	0.26 [0.20, 0.32]	0.07 [0.02, 0.13]	0.24 [20, 0.28]	0.10 [0.06, 0.15]	0.45 [0.40, 0.50]	–	
7. Verbal aggression	0.24 [0.18, 0.30]	0.13 [0.08, 0.18]	0.23 [0.19, 0.26]	0.05 [−0.05, 0.15]	0.30 [0.23, 0.38]	0.43 [0.36, 0.50]	–

Note. $M = 105$; $N = 36{,}215$.

3.3.2. Other forms of aggression

Given that there were so few studies that use measures of relational, sexual, or other forms of aggression, these correlations were meta-analyzed with multivariate meta-analysis. The top panel of Table 4 shows the correlations. There were not significant correlations between impulsivity and sexual aggression. For relational aggression, there were significant correlations (> 0.36) for negative and positive urgency and small, but nonsignificant correlations for the other impulsivity scales. There was a significant correlation for lack of premeditation and other aggression measures (e.g., indirect, driving). These results are generally in line with the main results but are based on fewer studies, so caution is warranted.

3.3.3. Functions of aggression

The bottom panel of Table 4 shows the zero-order and partial correlations using multivariate meta-analysis for reactive and proactive aggression. The zero-order correlations were significant for all facets of impulsivity and both aggression types ($r > 0.21$), aside from sensation seeking. For reactive aggression, the correlation was moderate-to-large for negative urgency and small-to-moderate for the other scales. For

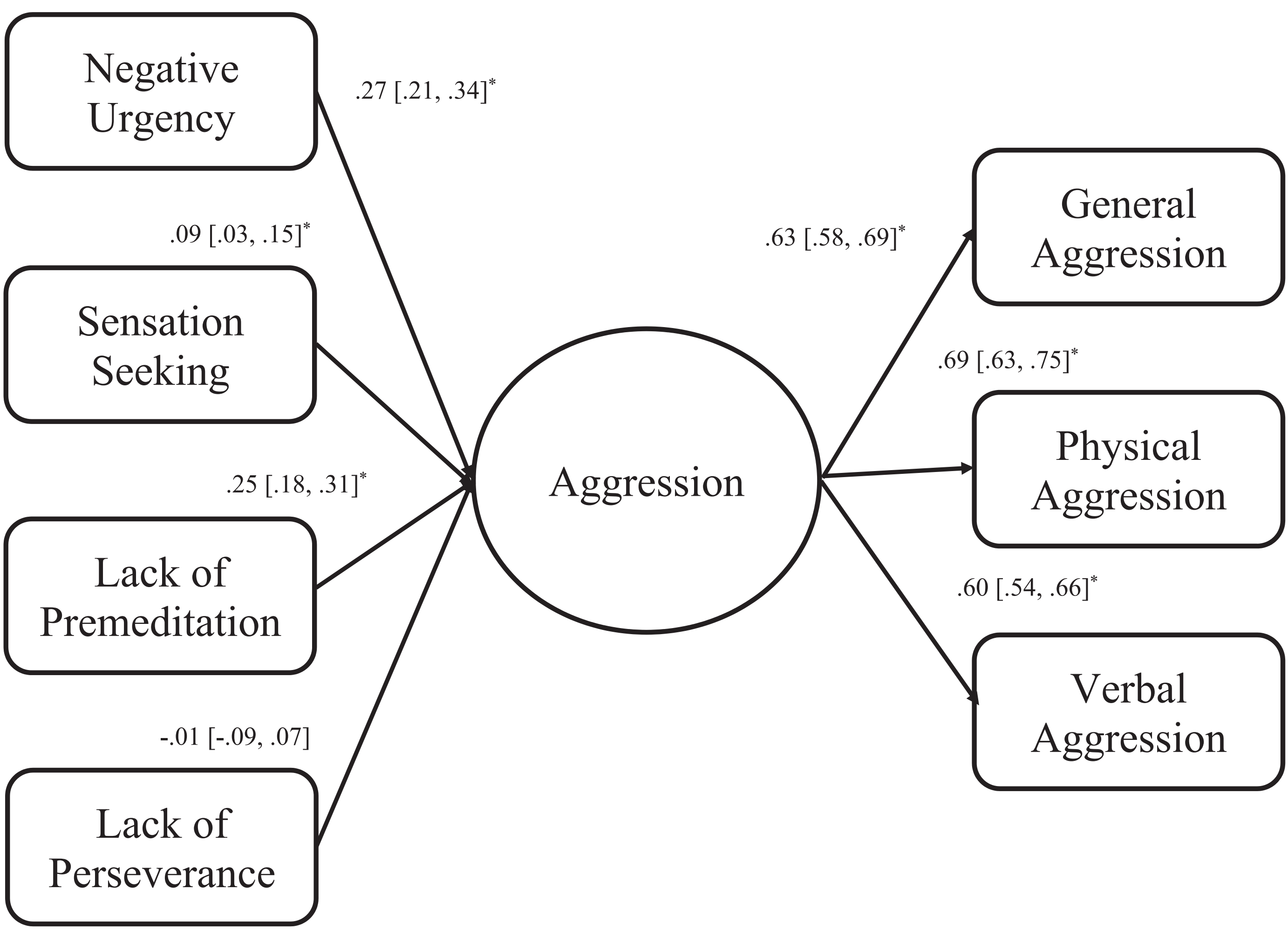

Fig. 2. Structural Equation Model for Impulsivity Predicting Aggression (Note Covariances Between Impulsivity Measures have been Omitted).

Table 4
Meta-analyzed Correlations and 95% Confidence Intervals Between Impulsivity and Secondary Aggression Outcomes.

	Sexual	Relational	Other
Negative urgency	−0.05, [−0.62, 0.52] k = 3, m = 2	0.36, [0.14, 0.57] k = 5, m = 3	0.24, [−0.11, 0.59] k = 3, m = 3
Positive urgency	–	0.37, [0.18, 0.55] k = 5, m = 3	–
Sensation seeking	–	0.11, [−0.94, 1.15] k = 4, m = 2	0.11, [−0.03, 0.24] k = 3, m = 3
Lack of premeditation	−0.00, [−0.92, 0.92] k = 3, m = 2	0.22, [−0.06, 0.49] k = 6, m = 3	0.17, [0.03, 0.32] k = 3, m = 3
Lack of perseverance	–	0.25, [−0.02, 0.52] k = 4, m = 2	0.12, [−1.43, 1.67] k = 2, m = 2

	Reactive aggression		Proactive aggression	
	Zero-order	Partial	Zero-order	Partial
Negative urgency (k = 5)	0.42 [0.34, 0.50]	0.32 [0.21,0.42]	0.29 [0.18, 0.39]	0.05 [−0.10, 0.19]
Positive urgency (k = 3)	0.30 [0.23, 0.37]	0.15 [0.05, 0.24]	0.41 [0.32, 0.50]	0.30 [0.18, 0.42]
Sensation seeking (k = 4)	0.23 [−0.01, 0.26]	0.21[−0.10, 0.24]	0.11 [−0.12, 0.32]	−0.04 [−0.33, 0.23]
Lack of premeditation (k = 6)	0.21 [0.11, 0.30]	0.03 [−0.01, 0.23]	0.31 [0.20, 0.42]	0.24 [0.17, 0.45]
Lack of perseverance (k = 6)	0.21 [0.09, 0.31]	0.02 [−0.13, 0.16]	0.32 [0.22, 0.41]	0.26 [0.12, 0.36]

Note. k = number of effects and m = number of studies (For the Bottom Panel k = m). Partial correlations were calculated with the meta-analyzed correlation between reactive and proactive aggression, r = 0.60, 95% CI [0.51, 0.67], k = 6.

proactive aggression, the correlation was numerically, but not significantly, largest for positive urgency.

For the unique variance of reactive aggression, these results showed that there were still significant correlations for negative and positive urgency (r > 0.15), but not any other scales (r < 0.03). For the unique variance of proactive aggression, the correlations were significant for positive urgency, lack of premeditation, and lack of perseverance (r > 0.24), but not sensation seeking or negative urgency. Together these results suggest a somewhat unique pattern of correlations for reactive aggression, which is associated with more reactive types of impulsivity, and proactive aggression which is associated with more cognitive aspects of impulsivity.

3.3.4. Demographic moderators

As detailed in the Supplementary Materials, sample age and sample gender composition were not significantly related to any effect sizes. There was a significant effect of sample type for lack of premeditation. The results showed that the correlation was stronger for community samples than undergraduate samples, but no other sample types were significantly different.

4. Discussion

The goal of this meta-analysis was to determine the relation between the unique facets of impulsivity and aggression. The results showed significant and small-to-medium correlations between each facet of impulsivity and aggression across several different forms of aggression. Consisting with my prediction negative urgency and lack of premeditation had significantly stronger associations with aggression than did sensation seeking and lack of perseverance. These results occurred both for zero-order correlations and when adjusting for overlap among impulsivity facets. Moreover, the results showed the correlation between different facets of impulsivity and aggression was consistent across forms of aggression (e.g., physical, verbal, general) and some evidence for a unique pattern for functions (i.e., reactive versus proactive) of aggression, at least when considering unique variance. These results help to elucidate the relation between different facets of impulsivity and aggression and can help improve theory in developmental, personality, social, and clinical psychology as well as other disciplines.

Consistent with theories of aggression (e.g., Anderson & Bushman, 2002; Verona & Bresin, 2015) all facets of impulsivity were significantly positively related to general and physical aggression. Moreover, all, but lack of perseverance, were significantly positivity related to verbal aggression. The results for other forms of aggression were more mixed, but these were based on fewer studies, which yielded imprecise effect size estimates. Together, these results suggest that in general, impulsive people are more likely to engage in aggressive behavior.

The results showed that the association was significantly stronger for negative urgency and lack of premeditation than sensation seeking and lack of perseverance. Although based on fewer studies, the results for positive urgency were similar in size to negative urgency. These results are in line with theories of aggression that highlight the importance of reactivity and inability to think about the consequences of behavior in aggression. Negative and positive urgency are conceptualized as impulsivity in the face of strong affect (Cyders & Smith, 2008) and thus capture the reactive nature of aggression. People high in these traits may be more reactive to aggression-instigating situations, and they may be less able to inhibit aggression when their emotional reactions are intense. Lack of premeditation is conceptualized as acting without planning (Whiteside & Lynam, 2001) and thus captures the lack of forethought aspect of aggression. Individuals high in lack of premeditation are less likely to think about the consequences of their aggressive-behavior and therefore are less likely to inhibit aggressive responding.

The effect sizes for negative urgency, positive urgency, and lack of premeditation would be considered small-to-medium in size by Cohen's (1992) standards and are in line with the median effect size in personality research, (Fraley & Marks, 2007). In comparison to other meta-analytic studies on correlates of aggression, the results for negative urgency, positive urgency, and lack of premeditation are larger. The effects for these three facets are larger than the effects for neighborhood disadvantage, academic achievement, 2d:4d digit ratio, and trait empathy (Chang, Wang, & Tsai, 2016; Savage, Ferguson, & Flores, 2017; Turanovic, Pratt, & Piquero, 2017; Vachon, Lynam, & Johnson, 2014) and are in line with personality traits like agreeableness and conscientiousness (Jones, Miller, & Lynam, 2011), suggesting that although the effects may be small-to-medium in size in comparison to effects in psychology, they large for individual differences in aggression research. Even though the effect sizes may be larger than other factors for aggression, there is still a large amount of variance unexplained by trait impulsivity. This fits with models of aggression that indicate important person-by-situation interactions (e.g., Finkel & Hall, 2018; Verona & Bresin, 2015). Future research may seek to understand under what conditions different aspects of impulsivity are particularly related to aggression.

Although the effect sizes for sensation seeking and lack of perseverance were significantly smaller than for the other UPPS facets, they were still significantly different from zero. Moreover, they were larger than some effects identified in prior meta-analyses as relating to aggression (e.g., neighborhood disadvantage, 2d:4d digit ratio; Chang et al., 2016; Turanovic et al., 2017), suggesting that they may still be important to consider in understanding aggression. The mechanism of how these facets are related to aggression remains unclear. It is possible that individuals high in sensation seeking are more likely to encounter aggression-provoking situations (e.g., their thrill-seeking adventures bring them into conflict with others). It is also possible that some individuals high in sensation seeking enjoy the thrill of aggressive behavior (e.g., being in a fight). Lack of perseverance may be particularly important for situations where individuals are trying to inhibit their aggression and need to persist against giving into the urge to aggress (e.g., a prolonged instigating situation). The fact that the association between lack of perseverance and aggression was not significant when adjusting for the overlap between other impulsivity scales may indicate that the association may have to do with the overlap of other factors. Future research could elucidate the nature of these associations.

For all facets of impulsivity, the results showed that the relation was consistent across forms (e.g., physical, verbal) of aggression and sample type. This suggests that impulsivity is associated with a general propensity to aggress, not necessarily more or less severe forms of aggression. This also indicates that impulsivity is not more or less related to aggression in more violent samples (e.g., incarcerated individuals) than less violent samples (e.g., undergraduates). Together, these analyses indicate that impulsivity facets are robust correlates of aggression.

The analyses looking at functions of aggression showed some unique patterns, particularly when considering the unique variance of reactive and proactive aggression. As predicted negative and positive urgency were significantly related to the unique variance of reactive aggression, whereas the other facets of impulsivity were not. This fits with the conceptualization of urgency as being a reactive form of impulsivity and suggests that individuals high in urgency may be most likely to aggress when provoked. This also in line with an instigator x disinhibitor interaction in I^3 theory (Finkel & Hall, 2018), with urgency playing the role of disinhibitor. Although more work is needed to understand the longitudinal relation, these results may suggest the individuals high in urgency may be a useful target for secondary prevention efforts to reduce reactive aggression. Furthermore, they may indicate that consistent with empirically-supported interventions for aggression (e.g., Lee & DiGiuseppe, 2018), reducing emotional responses to instigating situations would be an ideal treatment target.

The unique variance of proactive aggression was significantly positively related to lack of premeditation, lack of perseverance, and positive urgency. These results were not predicted but may be understood in the context of proactive aggression as working toward a goal. By definition, proactive aggression means using aggression for some other goal (e.g., obtain money) than to inflict pain. People who are good at planning ahead may be able to use means other than aggression to obtain their goal, but people who lack planning my turn to the less planful method of taking it by force. Similarly, people who can persist may suppress the urge to take a desirable object from someone, whereas people who lack perseverance may give into temptation and take it aggressively. Positive urgency may be related to proactive aggression through positive affect and approach motivation. Research and theory suggest that anger is an approach-related emotion (Carver & Harmon-Jones, 2009). In particular, anger is suggested to be a response to approach behavior not going well (e.g., a blocked goal; Carver & Harmon-Jones, 2009). Given that positive urgency is related to other approach emotions (i.e., positive affect) and is related to appetitive impulsive behaviors (e.g., risky sex, gambling; Cyders et al., 2007; Zapolski, Cyders, & Smith, 2009), it may be that when pursuit of goals is blocked, individuals high in positive urgency turn to aggression to pursue their goals. It should be noted that the majority of the studies used in the meta-analysis did not use the reactive proactive distinction; thus, these results were based on very few studies, and more work in this area is needed to understand these relations.

4.1. Limitations & strengths

These results should be considered in light of the limitations of this meta-analysis. First, although it is the best-known model of impulsivity, the UPPS is not the only way to organize facets of impulsivity (e.g., Johnson, Carver, & Joorman, 2013). It is possible that different results could emerge from a different model. Second, there were a large number of studies that included measures that made them eligible for this meta-analysis, but I was not able to obtain the necessary correlations. Therefore, this represents a somewhat limited sample of conducted studies. This is likely a function of the fact that many of these studies were not focused on impulsivity and aggression, yet included those measures, and thus did not report the relevant correlations in the text, rather than systematic non-reporting of null results. Still, this highlights the value in using supplemental materials and online repositories to make data available for future meta-analysts. Third, the majority of these studies used self-reported aggression. Further work with laboratory behavioral tasks and use of informants would provide collateral support for these results. Finally, these results are cross-sectional making it unclear whether impulsivity is a risk factor for aggression or vice versa. Longitudinal studies in this area are sorely needed to understand the validity of impulsivity in predicting future aggression.

This meta-analysis also has a number of strengths. Because of the relatively large number of studies and participant's per study, the confidence intervals were fairly narrow, yielding precise estimates of the relation between impulsivity and aggression. This allowed me to move beyond statistical significance and differentiate between medium and small effect sizes. These estimates can be used for sample size planning for future research and can also be used as benchmarks in the development of quantitative theories of aggression and/or impulsivity. This study used two-stage meta-analytic structural equation modeling, which was used to teste zero-order and partial relations. This helped rule out that smaller correlations were purely due to overlap with other aspects of impulsivity. This meta-analysis also used a broad set of studies that varied in many important characteristics (e.g., age, country, psychopathology). This suggests that these results are fairly generalizable.

The findings that negative urgency, positive urgency, and lack of premeditation have small-to-medium associations with aggression and sensation seeking and lack of perseverance have small associations can have implications for many areas of psychology. Developmental researchers may seek to understand the relation between different facets of impulsivity and aggression over the lifespan. Changes in these traits may explain different patterns of violence associated with age (e.g., Brame, Nagin, & Tremblay, 2001; O'Leary & Woodin, 2005). Social and personality psychologist may wish to understand how different aspects of trait impulsivity interact with situational factors to elicit or inhibit aggression. Clinical psychologists may use the stronger correlations for urgency and lack of premeditation to refine interventions for aggression by targeting emotional reactivity and identifying consequences of aggressive behavior. Forensic psychologists may extend this work to see how different aspects of impulsivity predict violent recidivism. These results also may be of interest to those outside of psychology (e.g., anthropology, public health, pharmacology) who study aggression. In general, these results help advance the field of aggression research by identifying impulsivity as involved in the pathophysiology of aggression.

Declaration of competing interest

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.avb.2019.08.003.

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